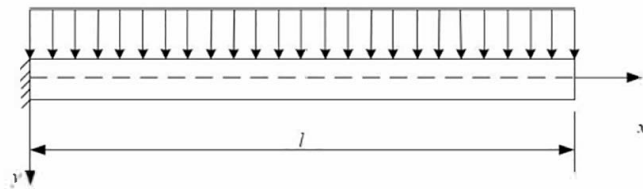


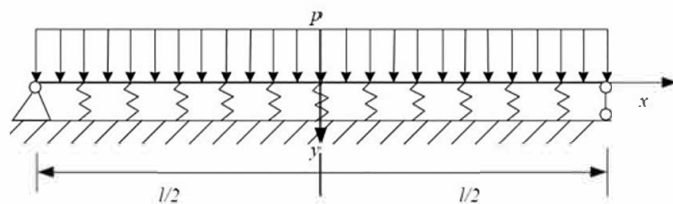
Homework 6

Submission date: 9th June, 2019

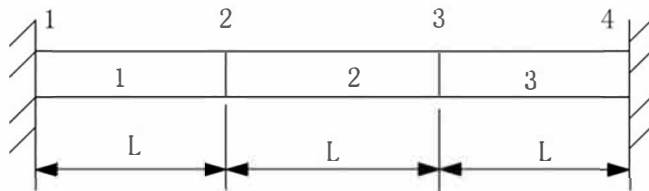
1. Prove that $w = a \left(1 - \cos \frac{\pi x}{2l} \right)$ is not the trial function of the Galerkin method for the cantilever beam below.



2. Try to obtain the differential equation and boundary condition of this elastic foundation beam by the variation method. And then solve it using first order approximation of Galerkin method.



3. The elastic rod is fixed at both ends. Suppose the section area is constant, the length is $3L$ and a body force f is uniformly applied. Divide the rod into 3 linear elements with the length of L and solve the element shape function matrix, element stiffness matrix and the global stiffness matrix.



4. Solve the following equations by the method of finite element. It is required to divide the interval into four elements by the nodes 0, 1/4, 1/2, 3/4 and 1.

$$\begin{cases} -u''(x) + u(x) = -x \\ u(0) = u(1) = 0 \end{cases}$$

Hints: The equivalent integral weak form of the equation is

$$Q[u(x)] = \frac{1}{2} \int_0^1 [u'^2(x) + u^2(x) + 2xu(x)] dx$$

5. Derive the element stiffness matrix of the linear triangle element.
6. Derive the element stiffness matrix of the rectangle element shown below.

